



HEAVY METAL TOXICITY AND NUTRIENT ANTAGONISM IN
NIGERIAN COCOA (*Theobroma cacao* L.): SOIL–PLANT TRANSFER,
FOOD SAFETY RISK AND MULTIVARIATE ASSESSMENT IN ABIA
AND OYO STATES

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DOI: <https://doi.org/10.70382/caijaat.v12i1.022>

ABSTRACT

Heavy metal contamination and nutrient antagonism in cocoa ecosystems have become a critical environmental and food safety concern in tropical agricultural systems because of the increasing anthropogenic activities, prolonged agrochemical use, and declining soil quality maintenance. This study evaluated heavy metal toxicity, nutrient interactions, soil–plant transfer, and associated food safety risks in cocoa (*Theobroma cacao* L.) plantations across Abia State and Oyo State respectively. Soil and cocoa plant parts (roots, barks, leaves, beans, and pod husks) were analyzed for selected macro/micronutrients and heavy metals using standard analytical procedures. Pollution and ecological risk indices including Bioaccumulation Factor (BAF), Transfer Factor (TF), Contamination Factor (CF), Geoaccumulation Index (Igeo), Pollution Load Index (PLI), Daily Intake of Metals (DIM), and Hazard Quotient (HQ) were computed using internationally accepted background values and toxicological reference doses. Multivariate statistical analyses involving principal component analysis, hierarchical clustering, correlation matrix, and analysis of variance were further used to identify metal sources, transfer patterns, and nutrient antagonism within the cocoa ecosystem. Results revealed significant variations ($p < 0.05$) in metal accumulation among cocoa plant tissues and between the two states. Elevated Fe and Mn concentrations were associated with reduced Zn and Cu availability which indicates strong nutrient antagonistic interactions under acidic soil conditions. Bioaccumulation and transfer indices suggested differential mobility of metals from soil to edible cocoa tissues, while contamination indices indicated varying degrees of ecological enrichment across sampling locations. Although most hazard quotient values remained below critical thresholds, continuous accumulation of Pb and associated metals may

constitute long-term food safety concerns. The study exclusively highlights the need for sustainable soil management and continuous monitoring of heavy metals in Nigerian cocoa production systems.

Keywords: Cocoa (*Theobroma cacao* L.), Bioaccumulation Factor, Nutrient Antagonism, Heavy Metal Toxicity, Food Safety Risk, Multivariate Analysis, Nigeria.

INTRODUCTION

Heavy metal contamination in cocoa production systems has attracted increasing scientific attention because of concerns regarding environmental sustainability, food safety, and international trade regulations. Recent studies have shown that prolonged application of fertilizers, fungicides, pesticides, and other agrochemicals contributes significantly to the accumulation of trace metals in cocoa plantation soils (Adewoye & Amusa, 2021; Shittu et al., 2024; Yeboah et al., 2024). Copper-based fungicides remain one of the major anthropogenic sources of copper enrichment in cocoa ecosystems, particularly in long-established plantations (Adeyeye et al., 2005; Alloway, 2013). Investigations conducted in cocoa-growing regions of Ghana, Nigeria, Ecuador, and Peru have demonstrated measurable enrichment of Cu, Pb, Zn, Cd, and Mn in plantation soils, with contamination levels generally increasing with plantation age and intensity of agrochemical use (Yeboah et al., 2024; García Porras et al., 2025). Although concentrations often remain below critical environmental thresholds, long-term accumulation may alter soil quality, reduce microbial activity, and increase the risk of metal uptake by cocoa plants (Kabata-Pendias, 2011; Sparks, 2003).

Among the heavy metals investigated in this study, Iron (Fe) is one of the most abundant and environmentally significant trace elements in cocoa production systems. Although iron is an essential micro nutrient required for chlorophyll synthesis, electron transport, enzymatic activity, and plant respiration, excessive accumulation in soils can adversely affect nutrient availability and plant productivity (Alloway, 2013; Kabata-Pendias, 2011). Iron enters cocoa agroecosystems primarily through the weathering of iron-rich parent materials, prolonged application of fertilizers and agrochemical, irrigation water, and natural pedogenic processes. In the highly weathered tropical soils of cocoa-growing regions, iron is often present in elevated concentrations because of intense mineral weathering and oxidation processes (Kabata-Pendias, 2011; Sparks, 2003).

The mobility and bio-availability of iron in soils are strongly influenced by soil pH, redox potential, organic matter content, and interactions with other mineral elements. Under acidic conditions, which are common in tropical rainforest soils, iron becomes more soluble and readily available for plant uptake, whereas alkaline conditions generally reduce its availability through precipitation reactions (Lindsay & Norvell, 1978; McBride, 1994). Excessive iron accumulation may also induce nutrient antagonism by suppressing the uptake of essential micro nutrients such as zinc (Zn), copper (Cu), and manganese (Mn), thereby affecting plant growth, physiological functions, and crop productivity (Kabata-Pendias, 2011; Sparks, 2003). Recent studies have demonstrated that cocoa plants readily absorb iron from the soil and

redistribute it among different plant tissues, particularly roots, leaves, and beans, reflecting its vital role in plant metabolism and development (Yeboah et al., 2024; Queiroz et al., 2026). Consequently, understanding the dynamics of iron accumulation and translocation is essential for evaluating nutrient balance, soil fertility, and the long-term sustainability of cocoa production systems.

Lead (Pb) contamination remains another major concern because of its toxicity, persistence, and bio accumulation potential. Lead is generally retained in root tissues because of its low mobility within plants (McBride, 1994). Nevertheless, contamination of cocoa beans may occur through atmospheric deposition, post-harvest handling, drying processes, transportation, and industrial emissions (Anyimah-Ackah et al., 2019). Studies evaluating heavy metals in cocoa beans and chocolate products have consistently reported detectable concentrations of Pb, raising concerns regarding chronic dietary exposure among consumers (Anyimah-Ackah et al., 2019; Förste et al., 2023). Continuous exposure to toxic metals may result in neurological disorders, renal dysfunction, carcinogenic effects, and developmental abnormalities, particularly among vulnerable populations (WHO, 2007; USEPA, 2011). The concept of soil–plant transfer has become increasingly important in understanding heavy metal dynamics within tropical agroecosystems. Soil–plant transfer factors (TF) and bio accumulation factors (BAF) are commonly used to quantify the movement of metals from soils into roots and subsequently into edible plant tissues (Håkanson, 1980; Tomlinson et al., 1980). Numerous studies have demonstrated that transfer efficiency varies considerably among metals and plant species. Cadmium generally exhibits relatively high mobility and transfer potential, whereas lead tends to accumulate predominantly in root systems (García Porras et al., 2025; Queiroz et al., 2026). The extent of transfer is strongly influenced by soil pH, cation exchange capacity, organic matter content, and competing ions (Kabata-Pendias, 2011; Lindsay & Norvell, 1978). Tropical soils characterized by intense weathering and acidic conditions often promote increased metal solubility and plant availability (Adriano, 2001; McBride, 1994). Consequently, transfer factor studies provide valuable information for predicting food-chain contamination and developing sustainable management strategies (Queiroz et al., 2026).

Human health risk assessment has become an integral component of heavy metal studies in agricultural systems. Indices such as Daily Intake of Metals (DIM), Estimated Daily Intake (EDI), Hazard Quotient (HQ), Hazard Index (HI), and Incremental Lifetime Cancer Risk (ILCR) are widely employed to evaluate potential health effects associated with contaminated food consumption (USEPA, 2011; WHO, 2007). Hazard Quotient values below unity generally indicate negligible non-carcinogenic risk, whereas values greater than one suggest potential adverse health effects (USEPA, 2011). Recent studies involving cocoa products and other agricultural commodities have reported generally low HQ values for essential elements but elevated risks where cadmium and lead concentrations exceed permissible limits (Anyimah-Ackah et al., 2019; Yeboah et al., 2024). These approaches provide quantitative frameworks for assessing consumer safety and supporting evidence-based environmental policies.

Multivariate statistical techniques have become indispensable tools in environmental contamination studies. Principal Component Analysis (PCA), Hierarchical Cluster Analysis (HCA), Factor Analysis

(FA), and correlation analysis are frequently employed to identify contamination sources, distinguish natural from anthropogenic inputs, and evaluate relationships among soil properties and heavy metals (Bhuiyan et al., 2010; Omeke & Igwe, 2021). PCA is particularly useful for reducing data dimensionality and identifying dominant factors controlling heavy metal distribution, whereas HCA facilitates grouping of sampling locations according to contamination characteristics and environmental similarity (Bhuiyan et al., 2010). Recent investigations have demonstrated the effectiveness of integrating pollution indices with multivariate statistical methods to improve interpretation of complex environmental datasets and identify major drivers of heavy metal accumulation in agricultural ecosystems (Omeke & Igwe, 2021; Olim et al., 2025).

Nutrient antagonism represents another critical aspect of heavy metal behavior in cocoa production systems. Excessive concentrations of iron (Fe), manganese (Mn), and copper (Cu) may suppress the uptake of zinc (Zn), calcium (Ca), magnesium (Mg), and potassium (K) through competitive interactions at root uptake sites (Kabata-Pendias, 2011; McBride, 1994). High Fe and Mn concentrations are particularly common in acidic tropical soils and may reduce zinc bio availability through precipitation reactions and competitive absorption mechanisms (Lindsay & Norvell, 1978; Sparks, 2003). Similarly, elevated copper concentrations resulting from repeated fungicide applications may interfere with zinc transport and physiological utilization, thereby contributing to nutrient imbalances and reduced crop productivity (Adeyeye et al., 2005; Alloway, 2013). Contemporary research emphasizes the importance of integrated nutrient management approaches that simultaneously address nutrient deficiencies, heavy metal toxicity, and soil health restoration in tropical agricultural systems (Queiroz et al., 2026; Yeboah et al., 2024).

Despite the increasing body of research on heavy metal contamination in cocoa production systems, comparative environmental assessments across Nigeria's major cocoa-producing ecological zones remain limited. In particular, little information exists regarding how the contrasting environmental characteristics of the southeastern rainforest zone (Abia State) and the southwestern rainforest zone (Oyo State) influence nutrient dynamics, heavy metal accumulation, and soil–plant transfer processes in cocoa ecosystems. These regions differ considerably in their geological parent materials, soil mineralogy, weathering intensity, and long-term land-use histories, all of which govern the distribution, mobility, and bio availability of nutrients and trace metals. The predominantly Coastal Plain Sands and sedimentary formations underlying much of southeastern Nigeria generally produce highly leached, acidic soils with mineralogical characteristics that differ from the Basement Complex-derived soils common in southwestern Nigeria. Furthermore, cocoa plantations in the two regions have experienced varying histories of fertilizer application, fungicide use, and general agronomic management, resulting in differences in soil chemical properties and potential heavy metal inputs. Such variations may influence nutrient antagonism, metal bio accumulation, ecological risk, and food safety. Therefore, a comparative assessment of cocoa production systems in Abia and Oyo States is essential to identify region-specific patterns of heavy metal behavior, improve understanding of soil–plant interactions, and provide scientific

evidence for developing sustainable nutrient management strategies and environmental monitoring programmes that enhance cocoa productivity while safeguarding consumer health.

Collectively, these studies highlight the importance of integrated assessment of heavy metal contamination, nutrient interactions, soil–plant transfer dynamics, and food safety risks in cocoa agro ecosystems. Such approaches are essential for developing sustainable management strategies that enhance cocoa productivity while ensuring environmental quality and consumer safety.

Aim of the Study

To evaluate nutrient dynamics, heavy metal accumulation, soil–plant transfer relationships, and ecological risks in cocoa production systems of Abia and Oyo States, Nigeria.

Specific Objectives are:

- To determine and compare the concentrations of macro-nutrients and heavy metals in cocoa plantation soils and different cocoa plant tissues across cocoa ecosystems in Abia and Oyo States.
- To evaluate the soil–plant transfer, bio accumulation, and interactions of nutrients and heavy metals, including the influence of soil properties on metal mobility, nutrient availability, and food safety risks in cocoa beans.
- To characterize and classify cocoa production environments using multivariate statistical techniques based on nutrient dynamics and heavy metal distribution patterns.

METHODOLOGY

Study Area

The study was conducted in selected cocoa-producing areas of Abia State (Umuahia located at approximately 5°31'N, 7°29'E) and Oyo State (Ibadan is located at approximately 7°23'N, 3°56'E), representing the rainforest agro ecological zones of southeastern and southwestern Nigeria, respectively. A total of 10 cocoa plantations were included in the study, comprising 5 plantations in each state. Plantations were selected using a simple random sampling approach from a list of actively cultivated cocoa farms and plots of Cocoa Research institute of Nigeria in both States. To minimize sampling bias, each plantation was considered an independent sampling unit. At each sampling point, mature and apparently healthy cocoa trees were randomly selected, and samples of roots, bark, leaves, beans, and pod husks were collected. The three field replicates from each plantation were composited to obtain representative samples for laboratory analyses. In total, each analytical determination was performed using three independent replicate samples, and all laboratory analyses were conducted in triplicate to improve analytical reliability. These regions are characterized by intensive cocoa cultivation, high annual rainfall, humid tropical climate, and prolonged use of agrochemicals associated with cocoa production systems. The soils of the study areas are predominantly highly weathered tropical soils susceptible to nutrient depletion, acidification, and heavy metal accumulation.

Composite rhizosphere soil samples were collected from cocoa plantations at a depth of 0–15 cm using a stainless-steel soil auger. Soil samples collected from multiple points within each sampling location were homogenized to obtain representative composite samples for laboratory analyses.

Cocoa plant samples comprising roots, bark, leaves, beans, and pod husks were collected from mature and apparently healthy cocoa plants within the sampled plantations. The collected plant samples were carefully washed with distilled water to remove adhered soil particles and contaminants. Samples were thereafter oven-dried at 65–70°C to constant weight, pulverized using a laboratory grinder, and stored in labeled airtight polyethylene containers prior to chemical analyses. The nutrient and heavy metal concentration values obtained from the analyzed soils and cocoa plant tissues are presented in Tables 1–3.

Determination of Soil Physicochemical Properties

Soil pH was determined potentiometrically using a glass electrode pH meter in a soil-to-water suspension ratio of 1:2.5. Organic carbon was determined using the Walkley–Black wet oxidation method, while total nitrogen was analyzed using the Kjeldahl digestion procedure. Available phosphorus was extracted using the Bray-1 extraction method.

Exchangeable potassium (K) and sodium (Na) were determined using a flame photometer, whereas calcium (Ca) and magnesium (Mg) were analyzed using Atomic Absorption Spectrophotometry (AAS) calibrated using multi-element standard solutions prepared from certified 1000 mg L⁻¹ stock standards. Soil cation exchange capacity (CEC) was determined by the ammonium acetate saturation method.

Determination of Nutrient Concentrations in Cocoa Tissues

Macro-nutrients analyzed in cocoa plant tissues included: Available phosphorus (P), Nitrogen (N), Carbon (C), Potassium (K), Calcium (Ca), Magnesium (Mg).

Plant tissue samples were subjected to wet acid digestion prior to analysis. Nitrogen concentration was determined using the Kjeldahl method, phosphorus was analyzed colorimetrically, while K, Ca, and Mg were determined using flame photometry and Atomic Absorption Spectrophotometry, respectively.

The nutrient concentrations in roots, bark, leaves, beans, and pod husks of cocoa plants from Abia and Oyo States are presented in Table 1.

Heavy Metal Analysis:

Heavy metals analyzed in soils and cocoa plant tissues included: Copper (Cu), Iron (Fe), Zinc (Zn), Lead (Pb), Manganese (Mn).

Soil and plant samples were digested using a tri-acid digestion mixture consisting of nitric acid (HNO₃), perchloric acid (HClO₄), and sulfuric acid (H₂SO₄). Concentrations of heavy metals were subsequently determined using Atomic Absorption Spectrophotometry (AAS).

The concentrations of heavy metals detected in cocoa plant tissues from both study areas are presented in Table 2, while soil-root-bean relationships are shown in Table 3.

Table 1: Micro and Macro nutrients in parts of Cocoa plant

Nutrients	State	Roots	Barks	Leaves	Beans	Pod husks
AvaiP(mg/kg)	Abia	34.8	20.85	30.16	28.77	29.55
AvaiP(mg/kg)	Oyo	31.18	16.36	22.69	23.95	27.75
%Nitrogen	Abia	0.3	0.14	0.35	0.31	0.51
%Nitrogen	Oyo	0.25	0.09	0.33	0.31	0.36
%Carbon	Abia	3.53	2.26	2.56	2.21	2.94
%Carbon	Oyo	3.28	1.64	3.35	2.44	3.96
K ⁺ (Cmol/kg)	Abia	0.44	0.38	1.14	2.49	6.99
K ⁺ (Cmol/kg)	Oyo	0.55	0.43	1.02	2.36	3.5
Ca ²⁺ (Cmol/kg)	Abia	2.73	3.12	31.44	21.35	26.63
Ca ²⁺ (Cmol/kg)	Oyo	2.86	4.31	19.33	18.14	18.48
Mg ²⁺ (Cmol/kg)	Abia	1.99	2.61	2.62	4.87	3.35
Mg ²⁺ (Cmol/kg)	Oyo	0.61	0.84	1.76	4.75	4.17

Table 2: Levels of Heavy Metals in parts of Cocoa plant

Heavy metal(mg/kg)	State	Roots	Barks	Leaves	Beans	Pod husks
Cu ²⁺	Abia	0.27	0.14	0.25	0.38	0.19
Cu ²⁺	Oyo	0.24	0.12	0.19	0.36	0.18
Fe ²⁺	Abia	23.56	2.04	12.31	30.3	5.78
Fe ²⁺	Oyo	21.46	3.45	12.38	23.78	5.86
Zn ²⁺	Abia	2.51	1.38	2.38	3.15	1.69
Zn ²⁺	Oyo	2.27	1.34	2.24	3.1	1.64
Pb ²⁺	Abia	0.42	0.21	0.18	0.15	0.24
Pb ²⁺	Oyo	0.68	0.22	0.43	0.07	0.35
Mn ²⁺	Abia	2.77	1.56	7.98	9.15	6.02
Mn ²⁺	Oyo	2.92	1.61	6.55	3.34	4.86

Table 3: Levels of nutrients and Heavy metals in Cocoa roots and Beans

Nutrients	State	Depths(0-15)cm	Roots	Beans
P(mg/kg)	Abia	37.8	34.8	28.77
P(mg/kg)	Oyo	35.43	31.18	30.18
%Carbon	Abia	4.18	3.53	2.01
%Carbon	Oyo	3.93	3.29	2.44
%Nitrogen	Abia	0.34	0.3	0.31

Nutrients	State	Depths(0-15)cm	Roots	Beans
%Nitrogen	Oyo	0.31	0.25	0.24
K ⁺ (Cmol/kg)	Abia	0.08	0.44	2.49
K ⁺ (Cmol/kg)	Oyo	0.07	0.55	2.35
Ca ²⁺ (Cmol/kg)	Abia	24.68	2.75	21.37
Ca ²⁺ (Cmol/kg)	Oyo	20.62	2.86	18.14
Mg ²⁺ (Cmol/kg)	Abia	7.77	1.99	4.87
Mg ²⁺ (Cmol/kg)	Oyo	3	0.61	4.75
Cu ²⁺ (Cmol/kg)	Abia	0.94	0.27	0.38
Cu ²⁺ (Cmol/kg)	Oyo	0.47	0.24	0.36
Fe ²⁺ (mg/kg)	Abia	22.71	23.54	30.31
Fe ²⁺ (mg/kg)	Oyo	14.33	21.46	23.78
Zn ²⁺ (mg/kg)	Abia	5.69	2.5	3.15
Zn ²⁺ (mg/kg)	Oyo	2.63	2.27	3.11
Pb ²⁺ (mg/kg)	Abia	0.3	0.41	0.15
Pb ²⁺ (mg/kg)	Oyo	0.31	0.68	0.07
Mn ²⁺ (mg/kg)	Abia	10.7	2.77	9.15
Mn ²⁺ (mg/kg)	Oyo	13.12	5.83	3.23

Soil–Plant Transfer Assessment

The movement and accumulation of heavy metals from soils into cocoa tissues were evaluated using bioaccumulation and transfer indices.

Bioaccumulation Factor (BAF)

$$BAF = C_{\text{plant}} / C_{\text{soil}} \dots\dots\dots(1.0)$$

Where:

C_{plant} = concentration of heavy metal in cocoa tissue,

C_{soil} = concentration of heavy metal in soil.

The BAF was used to determine the accumulation efficiency of metals from soils into cocoa plant tissues.

Transfer Factor (TF)

$$TF = C_{\text{bean}} / C_{\text{root}} \dots\dots\dots(1.1)$$

where:

C_{bean} = concentration of heavy metal in cocoa beans,

C_{root} = concentration of heavy metal in cocoa roots.

The transfer factor was used to assess translocation of heavy metals from roots into edible cocoa beans.

Pollution and Food Safety Assessment

To evaluate contamination status and food safety implications, pollution indices and health risk indicators were computed.

Contamination Factor (CF)

$$CF = C_{\text{sample}} / C_{\text{background}} \dots \dots \dots (1.2)$$

where:

C_{sample} = measured metal concentration,

$C_{\text{background}}$ = background concentration of the metal.

Pollution Load Index (PLI)

$$PLI = (CF_1 \times CF_2 \times \dots \times CF_n)^{1/n} \dots \dots \dots (1.3)$$

The PLI was used to evaluate the overall level of heavy metal pollution in cocoa soils.

Geoaccumulation Index (Igeo)

$$I_{\text{geo}} = \log_2 \left(\frac{C_n}{1.5B_n} \right) \dots \dots \dots (1.4)$$

Assesses anthropogenic contribution to heavy metal contamination.

Daily Intake of Metals (DIM)

$$DIM = C_{\text{metal}} \times D_{\text{food}} / \text{Body Weight} \dots \dots \dots (1.5)$$

The DIM was used to estimate human exposure to heavy metals through cocoa consumption.

Hazard Quotient (HQ)

$$HQ = \frac{DIM}{RfD} \dots \dots \dots (1.6)$$

where:

RfD = oral reference dose.

HQ values greater than 1 indicate potential non-carcinogenic health risks.

Statistical Analysis

Data generated from soil and plant were subjected to descriptive and inferential statistical analyses using R statistical software.

Descriptive statistics including: mean, range, standard deviation, coefficient of variation, were computed for all measured variables.

Analysis of variance (ANOVA) was performed to determine significant differences in nutrient and heavy metal concentrations among cocoa plant tissues and between the two states. Means were separated using Tukey's Honest Significant Difference (HSD) test at 5% probability level.

Pearson correlation analysis was used to evaluate relationships among nutrients and heavy metals and to determine possible antagonistic and synergistic interactions.

Multivariate statistical analyses including Principal Component Analysis (PCA), Hierarchical Cluster Analysis (HCA), correlation matrix analysis, and heatmap visualization were further employed to identify major controlling variables, classify sampling units, and explain patterns of heavy metal accumulation and nutrient interactions within the cocoa ecosystems.

Background concentrations used for contamination assessment and geoaccumulation analyses were adopted from the average shale values reported by Turekian and Wedepohl (1961), which are widely applied in environmental geochemistry studies.

Average Shale Background Values

Metal	Background (mg/kg)
Pb	20
Cu	45
Zn	95
Fe	47200
Mn	850

Source: Turekian, K.K. and Wedepohl, K.H. (1961). Distribution of the elements in some major units of the earth's crust.

The Reference Dose (RfD) values used were adopted from the oral Reference Dose database of the United States Environmental Protection Agency Integrated Risk Information System (IRIS).

Heavy Metal	RfD (mg/kg/day)	Primary Source
Pb	0.0035	USEPA / WHO provisional guideline
Cu	0.04	USEPA IRIS
Zn	0.30	USEPA IRIS
Fe	0.70	USEPA / ATSDR nutritional threshold
Mn	0.14	USEPA IRIS

Method Detection Limits (MDLs) were established as three times the standard deviation (3σ) of repeated reagent blank measurements, in accordance with the United States Environmental Protection Agency (USEPA) procedure (USEPA, 2016). The detection limits achieved for the analyzed heavy metals were as follows:

Heavy metal	Detection limit
Copper (Cu)	0.01 mg kg ⁻¹
Iron (Fe)	0.10 mg kg ⁻¹
Zinc (Zn)	0.01 mg kg ⁻¹
Lead (Pb)	0.005 mg kg ⁻¹
Manganese (Mn)	0.01 mg kg ⁻¹

All measured concentrations exceeded their respective detection limits, confirming that the analytical method possessed adequate sensitivity for reliable quantification of heavy metals in both soil and cocoa plant tissues.

- All analysis were performed in the R environment.

RESULTS AND DISCUSSION

The descriptive statistics provided Mean, Standard Deviation (SD), Minimum, Maximum, and Coefficient of Variation (CV).

Carbon (%C)

Highest carbon concentration occurred in: Roots (3.73–3.85%), Pod husks (2.94–3.96%). This indicates stronger organic matter accumulation in underground tissues and cocoa residues (pod husks). Higher carbon in pod husks suggests they can contribute significantly to soil organic matter recycling if returned to the farm. Roots also showed low CV (~12%) and Barks showed higher variability (31–47%) This suggests carbon distribution was more stable in roots than in bark tissues.

Nitrogen (%N)

Pod husks had the highest nitrogen concentration (0.435%) while Bark had the lowest (0.195%). High nitrogen in pod husks suggests strong nutrient re-mobilization during fruit development while low nitrogen in bark indicates lower metabolic demand in woody tissues.

Potassium (K^+)

Pod husks had extremely high K concentration (5.245 Cmol/kg) while Roots had the lowest (0.285 Cmol/kg). Potassium is highly mobile and strongly associated with fruit development and osmotic regulation. Cocoa pod husks act as major potassium sinks. This indicates that Cocoa harvesting removes substantial K from farms and Continuous harvesting without K fertilization may deplete soil fertility. The very high CV (54.3%) indicates strong variability in potassium distribution among tissues.

Calcium (Ca^{2+})

Leaves and pod husks recorded the highest concentrations (~22.5 Cmol/kg) while Bark had the lowest (~3.26 Cmol/kg). Calcium accumulation in leaves reflects its structural role in cell walls and membrane stability. Since calcium is relatively immobile in plants, it accumulates in transpiring tissues like leaves.

Magnesium (Mg^{2+})

Differences among plant parts were relatively small. This implies that Magnesium distribution was more uniform because Mg is centrally involved in chlorophyll synthesis and photosynthesis across tissues.

Heavy Metal Distribution

Copper (Cu^{2+})

Beans had the highest Cu concentration (0.37) while Bark had the lowest (0.13). Copper is an essential micro-nutrient involved in enzyme activation and electron transport. Higher Cu in beans suggests

translocation toward reproductive tissues. However, concentrations remained relatively low and within safe limits.

Iron (Fe^{2+})

Beans (27.04 mg/kg) and roots (22.51 mg/kg) showed highest Fe concentrations while Bark had the lowest (2.75 mg/kg). This is because Iron is required for chlorophyll synthesis and respiration. Roots accumulated Fe due to direct soil contact, while substantial movement into beans indicates active nutrient translocation. The high Fe in beans may improve nutritional value but excessive accumulation may become toxic if concentrations increase further.

Zinc (Zn^{2+})

Beans had highest Zn concentration (3.125 mg/kg) while the Bark had lowest concentration (1.36 mg/kg). Zinc is essential for enzyme systems and reproductive growth. Elevated Zn in beans indicates physiological transport toward seeds during maturation.

Lead (Pb^{2+})

Roots had highest Pb concentration (0.55 mg/kg) while Beans had lowest Pb concentration (0.11 mg/kg). Lead accumulation mainly in roots suggests restricted upward movement within the cocoa plant. This is environmentally beneficial because edible portions (beans) accumulated much lower Pb concentrations. This indicates cocoa plants possess partial exclusion mechanisms against Pb translocation.

Manganese (Mn^{2+})

Leaves had highest Mn concentration (7.27 mg/kg). Manganese participates in photosynthetic reactions, especially water splitting in photo system. High accumulation in leaves reflects its direct involvement in photosynthesis.

Two-Way ANOVA

The ANOVA evaluated effects of:

- i. Plant part
- ii. State
- iii. State $\times$ Plant part interaction

Significant Effect of Plant Part:

Potassium (K^+)

PlantPart: $p = 0.000304$, this indicates potassium concentration differed significantly among cocoa plant parts. Pod husks accumulated significantly more potassium than roots, bark, and leaves depicting nutrient partitioning within Cocoa plant.

Calcium (Ca^{2+}): $p = 0.0176$, Nitrogen: $p = 0.0232$

The values above suggest moderate differences among plant parts but not strong enough at the 5% significance level.

State Effects:

The States were not significant indicating that Cocoa nutrient/heavy metal accumulation patterns were generally similar between Abia and Oyo states due to Similar environmental conditions, Comparable soil chemistry, Similar agronomic practices etc.

Interaction Effects:

State $\times$ PlantPart interactions were non-significant.

This is because the way nutrients accumulated across plant tissues did not depend strongly on location. Correlation Matrix: the correlation matrix shown in *Figure 1*, depicts positive for all factors. The strongest being leaves vs pod husk (0.953) which implies that Nutrient accumulation in leaves strongly mirrors pod husk accumulation. Other Strong Correlations includes: Roots vs Barks (0.852) and Leaves vs Beans (0.871). This means that Nutrient transfer occurs systematically among plant organs, coordinated nutrient uptake and redistribution.

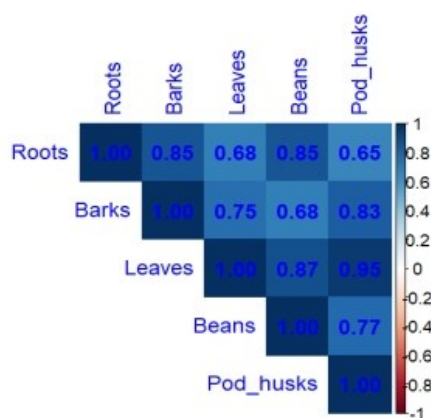


Figure 1: correlation Matrix of Cocoa plant parts

Principal Component Analysis (PCA):

PCA reduced dimensionality and identified major variation patterns. Variables were highly correlated. Plant tissues shared common nutrient accumulation trends. *Figures 2 and 3*.

PC1 represented overall nutrient/heavy metal accumulation while PC2 represented tissue-specific partitioning.

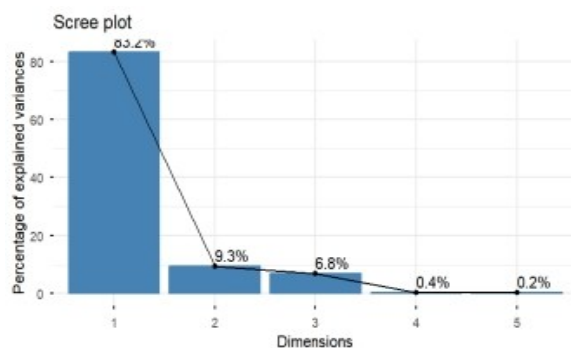


Figure 2: Scree plot of Components

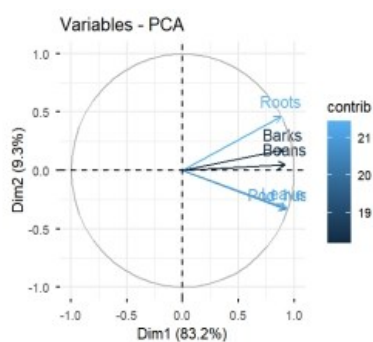


Figure 3: Biplot of Variables.

Roots and beans clustered together due to high Fe and Zn accumulation, while pod husks separated because of extremely high potassium content.

Hierarchical Cluster Analysis

The dendrogram, **Figure 4**, grouped similar nutrient profiles identifying three (3) major clusters implying that Some plant parts shared similar elemental composition and nutrient behavior followed physiological functions. Leaves and pod husks clustered together, Roots formed another cluster because of metal uptake functions.

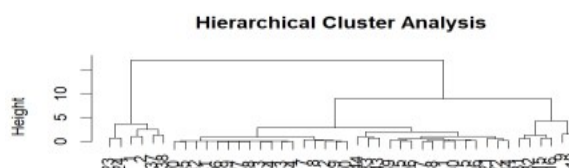


Figure 4: Dendrogram of Clusters

Heavy Metal Risk Assessment

Bioaccumulation Factor (BAF): **Figure 5**

All BAF values were < 1 indicating that Cocoa plants showed: Low metal accumulation ability, limited uptake from soil. Thus, cocoa behaved more as a metal excluder than hyperaccumulator.

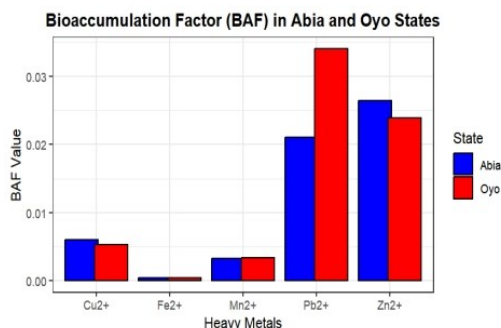


Figure 5: BAF plot for Abia and Oyo States.

Transfer Factor: **Figure 6**

Most TF values were < 1 as heavy metal movement from roots to beans was restricted.

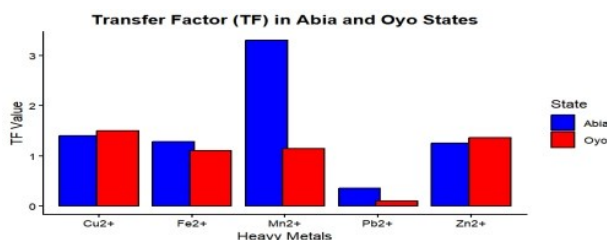


Figure 6: Transfer factor for Abia and Oyo States

Contamination Factor (CF): **Figure 7.**

CF values were approximately indicating that soils have low contamination relative to Turekian and Wedepohl background values therefore no serious anthropogenic pollution was evident and soil quality remained relatively safe.

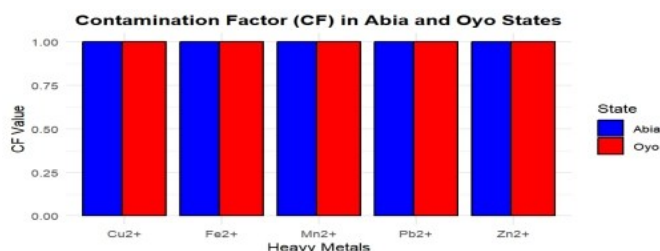


Figure 7: Contamination factors in Abia and Oyo States.

Geoaccumulation Index (Igeo)

Igeo values were below zero indicating that the soils were unpolluted and have minimal heavy metal enrichment.

Daily Intake of Metals (DIM): **Figure 8**

DIM values were very low depicting estimated human intake through cocoa consumption is minimal.

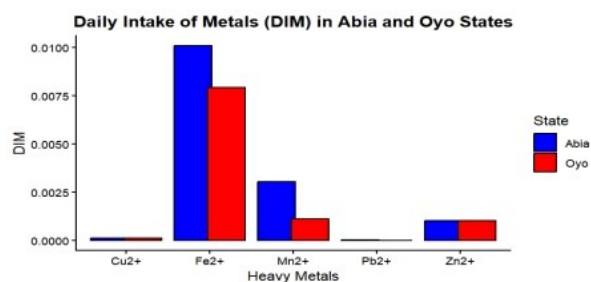


Figure 8: Daily Intake of Metal in Abia and Oyo States.

Hazard Quotient (HQ): **Figure 9**

All HQ values were < 1 indicating that no significant non-carcinogenic health risk exists from consuming cocoa beans from the study areas.

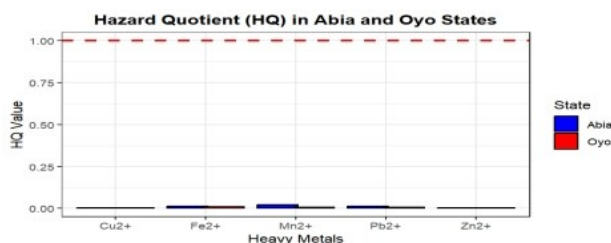


Figure 9: Hazard Quotient of Heavy metals in Abia and Oyo States.

This is a major food safety finding.

Pollution Load Index (PLI): **Figure 10**

PLI values were close to or below 1 depicting that the study soils were generally unpolluted.

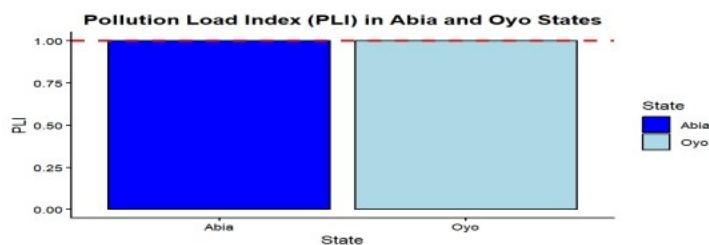


Figure 10: Pollution Load Index of Abia and Oyo States.

CONCLUSION

The present study demonstrates distinct patterns of nutrient and heavy metal partitioning in cocoa plants cultivated in Abia and Oyo States, highlighting the physiological mechanisms governing nutrient allocation and the environmental safety of cocoa production systems in these regions. The observed significant variation in elemental concentrations among leaves, beans, pod husks, and other plant tissues confirms that cocoa plants selectively distribute nutrients according to their functional roles in growth, reproduction, and metabolic activities. Such tissue-specific nutrient partitioning is a common characteristic of perennial crops and reflects differences in nutrient mobility, sink strength, and physiological demand during pod development and seed maturation. One of the most notable findings was the substantial accumulation of potassium (K) in cocoa pod husks. Potassium is highly mobile within plants and plays critical roles in osmoregulation, enzyme activation, carbohydrate translocation, and fruit development. The elevated K concentration in pod husks suggests that this tissue serves as an important reservoir for potassium during pod maturation. Similar observations have been reported in cocoa-producing regions where pod husks contain significantly higher K levels than other plant parts. This finding has important agronomic implications because cocoa pod husks represent a potentially valuable source of organic fertilizer. Recycling pod husks back into cocoa farms could enhance soil fertility, reduce dependence on inorganic fertilizers, and promote sustainable nutrient cycling within cocoa agro ecosystems. The study also revealed relatively high concentrations of iron (Fe) and zinc (Zn) in cocoa beans. Both elements are essential micro nutrients involved in numerous physiological processes. Iron participates in electron transport, chlorophyll synthesis, and enzymatic reactions, whereas zinc plays key roles in protein synthesis, enzyme activation, and hormone regulation. The preferential accumulation of Fe and Zn in cocoa beans may reflect the plant's strategy to supply developing seeds with essential micro nutrients required for germination and early seedling establishment. From a nutritional perspective, the enrichment of cocoa beans with Fe and Zn is particularly significant because cocoa-derived products contribute to dietary micro nutrient intake. Given the widespread occurrence of iron and zinc deficiencies in many developing countries, the presence of these micro nutrients in cocoa beans enhances the nutritional value of cocoa and its processed products. A major food safety finding was the restricted translocation of lead (Pb) from vegetative tissues into cocoa beans. Although trace amounts of Pb may be present in soils due to natural geological processes or anthropogenic activities, the low concentrations detected in edible tissues suggest the existence of physiological barriers that limit Pb mobility within the cocoa plant. Heavy metals such as Pb are often retained within root tissues or immobilized in cell walls and vacuoles, thereby reducing their transfer to reproductive organs. The limited accumulation of Pb in cocoa beans observed in this study is encouraging because lead is one of the most toxic environmental contaminants, with no known biological function in humans. These findings indicate that cocoa cultivated in the studied areas is unlikely to constitute a significant source of dietary Pb exposure. The strong positive correlations observed among plant tissues further suggest coordinated nutrient uptake, transport, and storage mechanisms. Positive interrelationships among leaves, pod husks, and beans

indicate that nutrient concentrations in one plant organ may influence or reflect concentrations in other organs. Such relationships may arise from common soil nutrient sources, synchronized physiological processes, or efficient internal nutrient redistribution. The strong associations among plant parts also imply that nutrient management practices targeting one tissue compartment could influence overall nutrient dynamics within the plant. Consequently, integrated nutrient management strategies may be particularly effective in optimizing cocoa productivity and quality. Environmental assessment indicators revealed minimal ecological contamination across the study locations. Low contamination levels indicate limited anthropogenic inputs of toxic metals and suggest that the agricultural soils supporting cocoa production remain relatively unpolluted. This observation is particularly important because increasing industrialization, mining activities, and indiscriminate agrochemical use have contributed to heavy metal accumulation in agricultural soils in many tropical regions. The low contamination status recorded in this study reflects favorable environmental conditions and supports the continued sustainability of cocoa cultivation in both Abia and Oyo States. The human health risk assessment further demonstrated that cocoa consumption from the investigated areas poses negligible risk to consumers. Hazard quotient (HQ) and hazard index (HI) values below the recommended safety threshold indicate that exposure to potentially toxic elements through cocoa consumption is unlikely to result in adverse health effects. These findings are consistent with the low concentrations of toxic metals detected in cocoa beans and reinforce the suitability of cocoa produced in these regions for both local consumption and international markets. Given the increasing global concern regarding heavy metal contamination of cocoa products, particularly cadmium and lead, the present results provide valuable evidence supporting the safety and quality of Nigerian cocoa. Overall, the study highlights the dual role of cocoa plants as efficient nutrient accumulators and selective barriers against toxic metal transfer. The predominance of beneficial nutrients such as potassium, iron, and zinc, coupled with the restricted accumulation of toxic metals in edible tissues, underscores the nutritional and economic value of cocoa produced in Abia and Oyo States. Furthermore, the low ecological contamination and negligible human health risks demonstrate that these cocoa production systems remain environmentally sustainable and suitable for long-term cultivation. Future studies should investigate seasonal variations, soil physicochemical factors, and genotype-dependent differences in nutrient and metal accumulation to further strengthen understanding of nutrient cycling and food safety within cocoa agro ecosystems.

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APPENDIX 1

Two-way ANOVA

Metal	Source	df	F-value	p-value
Cu²⁺	State	1	2815	<2×10 ⁻¹⁶
	Plant Part	5	7926	<2×10 ⁻¹⁶
	State × Plant Part	5	1470	<2×10 ⁻¹⁶
	Residual	24	—	—
Fe²⁺	State	1	1357.5	<2×10 ⁻¹⁶
	Plant Part	5	9657.5	<2×10 ⁻¹⁶
	State × Plant Part	5	384.6	<2×10 ⁻¹⁶
	Residual	24	—	—
Zn²⁺	State	1	2326	<2×10 ⁻¹⁶
	Plant Part	5	4306	<2×10 ⁻¹⁶
	State × Plant Part	5	1515	<2×10 ⁻¹⁶
	Residual	24	—	—
Pb²⁺	State	1	3938	<2×10 ⁻¹⁶
	Plant Part	5	6552	<2×10 ⁻¹⁶
	State × Plant Part	5	1468	<2×10 ⁻¹⁶
	Residual	24	—	—

Effect Sizes

Partial Eta Squared (η^2)

Metal	State	Plant Part	State × Plant Part
Cu²⁺	0.99	1.00	1.00
Fe²⁺	0.98	1.00	0.99
Zn²⁺	0.99	1.00	1.00

Partial Omega Squared (ω^2)

Metal	State	Plant Part	State × Plant Part
Cu²⁺	0.99	1.00	1.00
Fe²⁺	0.97	1.00	0.98
Zn²⁺	0.98	1.00	1.00

These indicate **very large effect sizes**, showing that State, Plant Part, and their interaction explain nearly all variation in metal concentrations.

Tukey HSD

Cu²⁺

State: Oyo – Abia = **-0.102**, $p < 0.001$

All Plant Part comparisons were significant ($p < 0.001$). Significant interaction comparisons were also observed for nearly all State × Plant Part combinations.

Fe²⁺

State: Oyo – Abia = **-2.853**, $p < 0.001$.

Beans contained significantly higher Fe than Barks. Roots, Soil, Leaves, and Pod Husks differed significantly from one another in most comparisons. Nearly all interaction comparisons were significant.

Zn²⁺

State: Oyo – Abia = **-0.608**, $p < 0.001$.

Nearly every Plant Part comparison was significant except Roots vs Leaves ($p = 0.0998$). Most interaction comparisons were significant.

Pb²⁺

State: Oyo – Abia = **0.092**, $p < 0.001$.

Most Plant Part comparisons were significant. Leaves vs Soil was not significant ($p = 0.779$). Most interaction comparisons were significant.

Component Loading

	Dim.1	Dim.2	Dim. 3	Dim. 4	Dim.5
Roots	0.9319	-0.3562	0.0118	0.0212	0.0641
Barks	0.9785	--0.0633	0.0883	-0.1735	-0.0262
Leaves	0.9878	0.1225	0.0675	0.0536	0.0401
Beans	0.9611	-0.2471	-0.0151	0.0967	-0.0759
Pod husks	0.9081	0.4092	0.0738	0.0486	-0.0086
Soil	0.9628	0.1440	-0.2249	-0.0415	0.0074